

JOSHUA OLIVEIRA

COMPUTER ENGINEERING GRADUATE

MAJOR GPA: 3.83

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✉️ California, USA

🏠 Website: [jraptor11.github.io/joshua-](https://jraptor11.github.io/joshua-oliveira-engineering-portfolio)

📁 [oliveira-engineering-portfolio](https://github.com/JRaptor11)

GitHub: github.com/JRaptor11

PROFESSIONAL SUMMARY

Computer Engineering graduate with experience designing, integrating, testing, and validating real-time embedded systems across autonomous and sensor-driven platforms. Skilled in Python and C/C++ development, hardware-software integration, root-cause analysis, and debugging complex systems in Linux environments. Experienced developing logging, monitoring, and validation workflows to analyze system behavior, identify failures, improve reliability, support fault isolation, and troubleshoot complex autonomous systems in real-time environments.

EXPERIENCE

Undergraduate Research Assistant — Thermal Control System

Loyola Marymount University | January 2025 – May 2025

- Engineered and tested a Raspberry Pi-based closed-loop thermal control system maintaining $\pm 0.5^{\circ}\text{C}$ stability using custom PCB instrumentation.
- Designed closed-loop control logic with real-time data acquisition and test validation workflows for stable operation under continuous load.
- Debugged performance using sensor feedback and resolved stability issues.

Teaching Assistant / Lab Assistant

Loyola Marymount University | January 2023 – May 2026

- Supported instruction across computer architecture, circuits, and mathematics, reinforcing systematic debugging and validation approaches.
- Guided 25+ students in troubleshooting electrical circuits, lab instruments, and test equipment, supporting signal analysis and system validation.

PROJECTS

Autonomous Driving Smart Car System — Autonomous Systems Lead

Loyola Marymount University | August 2025 – May 2026

- Architected and integrated a real-time embedded system on NVIDIA Jetson with CAN communication across sensing, compute, and control subsystems.
- Developed Python- and C++-based hardware-software interfaces, debugging data flow and performance issues, achieving <50 ms end-to-end latency.
- Executed system-level testing and validation to identify failure modes, verify functionality, and improve overall fault tolerance and system reliability.

Automated Data Processing & Monitoring System (Trading)

Independent Project | May 2025 – Present

- Designed and deployed a Python-based monitoring and automation system using FastAPI for real-time data processing in an event-driven architecture.
- Developed logging, monitoring, and validation pipelines to analyze runtime behavior, identify failures, and improve reliability under live conditions.
- Debugged and resolved runtime failures through root-cause analysis, implementing software fixes to improve system stability and reliability.

Discrete Op-Amp Design (Build & Test Lab)

Loyola Marymount University | January 2025 – May 2025

- Designed, built, and tested a discrete transistor-level operational amplifier with differential input, gain, and output stages.
- Applied feedback and compensation to stabilize frequency response, validating gain and bandwidth using oscilloscope measurements.
- Debugged circuit behavior and verified performance through lab testing and comparison to theoretical models, documenting measured results.

EDUCATION

B.S. in Computer Engineering

Minor in Applied Math

Minor in Business Administration

Loyola Marymount University (LMU)

Graduated May 2026

Engineering Honors Society

Tau Beta Pi – LMU Chapter

Engineering Study Abroad Program in Bonn, Germany through LMU

Academy for International Education

Completed Fall 2023

TECHNICAL SKILLS

Languages & Tools: Python, C/C++, VHDL, MATLAB, Assembly, Git/GitHub, Linux, NVIDIA Jetson, Raspberry Pi, Arduino, FastAPI, SolidWorks (CAD)

Systems: Embedded Systems, Real-Time Systems, Hardware-Software Integration, Sensor Integration

Interfaces: CAN, SPI, I2C, UART, USB

Validation: Oscilloscopes, Root Cause Analysis, Hardware Debugging, Failure Analysis, Diagnostics, Data Validation, Logging & Monitoring, Logic Analyzers

RELEVANT COURSEWORK

Computer Architecture with VHDL:

Explored instruction cycles, control logic, and pipelining to design robust hardware-software systems.

Microprocessors & Microcontrollers:

Programmed ARM Assembly to implement control and arithmetic logic while studying both low- and high-level processor architecture.

Electronic Design & Build Lab I & II:

Designed and tested analog and digital circuits using professional lab instrumentation and protocols.

Additional Coursework: Circuits I & II, Electronics, Signals & Linear Systems, Machine Learning, Data Structures & Algorithms, and Discrete Mathematics